## What is Loop?

In [ ]:
```
A Loop is used to execute the same block of code multiple times
until a condition becomes false.
```

In [1]:
```
print("Welcome")
print("Welcome")
print("Welcome")
print("Welcome")
print("Welcome")
```

```
Welcome
Welcome
Welcome
Welcome
Welcome
```

In [2]:
```
for i in range(5):
    print("Welcome")
```

```
Welcome
Welcome
Welcome
Welcome
Welcome
```

## Types of loops

In [ ]:
```
1)for Loop
2)while loop
```

## 1)for loop

In [ ]:
```
A for loop is used when we know how many times we want to repeat.
```

#### Syntax:

In [ ]:
```
for variable in range():
    statement
```

In [ ]:
```
for variable in sequence:
    statement
```

In [ ]:
```
for index,value in enumerate(sequence):
    statements
    print(index, value)
```

In [3]:
```
n=5
fact=1
for i in range(1,n+1):
    fact=fact*i # fact=1 =1*1=1 ,fact*2=2,fact*3=6
print(f"Factorial of {n} is {fact}")
```

```
Factorial of 5 is 120
```

In [4]:
```
n=5
fact=1

for i in range(5,0,-1):
    fact=fact*i
print(f"Factorial of {n} is {fact}")
```

```
Factorial of 5 is 120
```

In [5]:
```
n=7
fact=1
for i in range(n,0,-1):
    fact=fact*i
print(f"Factorial of {n} is {fact}")
```

```
Factorial of 7 is 5040
```

In [6]:
```
for i in [10,20,30,40,50]:
    print(i)
```

```
10
20
30
40
50
```

In [7]:
```
for i in 'python':     #P
    print(i.upper())
```

```
P
Y
T
H
O
N
```

## Loop Control Statements

In [ ]:
```
break
continue
pass
```

## 1.break

In [ ]:
```
It is used to terminate a for loop.
```

In [9]:
```
for i in 'data science':
    print(f'Char=={i}')
```

```
Char==d
Char==a
Char==t
Char==a
Char== 
Char==s
Char==c
Char==i
Char==e
Char==n
Char==c
Char==e
```

In [10]:
```
for char in 'Data Science':
    print(f"Char=={char}")
    if char.isupper(): #True
        print("Character is in upper case")
    print("*"*40)
```

```
Char==D
Character is in upper case
****************************************
Char==a
****************************************
Char==t
****************************************
Char==a
****************************************
Char== 
****************************************
Char==S
Character is in upper case
****************************************
Char==c
****************************************
Char==i
****************************************
Char==e
****************************************
Char==n
****************************************
Char==c
****************************************
Char==e
****************************************
```

In [11]:
```
for char in 'data Science':
    print(f"Char=={char}")
    if char.isupper(): #True
        print("Character is in upper case")
        break

    print("*"*40)
print("loop Completed")
```

```
Char==d
****************************************
Char==a
****************************************
Char==t
****************************************
Char==a
****************************************
Char== 
****************************************
Char==S
Character is in upper case
loop Completed
```

In [12]:
```
for char in 'data Science':
    print(f"Char=={char}")
    if char.isupper(): #True
        print("Character is in upper case")
        break
        print("Hello Python")

    print("*"*40)
print("loop Completed")
```

```
Char==d
****************************************
Char==a
****************************************
Char==t
****************************************
Char==a
****************************************
Char==
****************************************
Char==S
Character is in upper case
loop Completed
```

In [13]:
```
for char in 'data Science':
    print(f"Char=={char}")
    if char.isupper(): #True
        print("Character is in upper case")
        print("Hello Python")
        break

    print("*"*40)
print("loop Completed")
```

```
Char==d
****************************************
Char==a
****************************************
Char==t
****************************************
Char==a
****************************************
Char==
****************************************
Char==S
Character is in upper case
Hello Python
loop Completed
```

In [14]:
```
for i in range(5):
    print(f"Number=={i}")
```

```
Number==0
Number==1
Number==2
Number==3
Number==4
```

In [15]:
```
for i in range(5,10):
    print(f"Number=={i}")
    break
print("Loop completed")
```

```
Number==5
Loop completed
```

In [17]:
```
for i in range(5,11):
    print(f"Number=={i}")
    if(i==8)or(i==9):
        print("Number Found")
    print("*"*30)
print("loop completed")
```

```
Number==5
******************************
Number==6
******************************
Number==7
******************************
Number==8
Number Found
******************************
Number==9
Number Found
******************************
Number==10
******************************
loop completed
```

## 2. continue

In [ ]:
```
Used to stop current iteration and continue with next iteration.
```

In [18]:
```
for i in range(10,20):
    print(f"i=={i}")
```

```
i==10
i==11
i==12
i==13
i==14
i==15
i==16
i==17
i==18
i==19
```

In [19]:
```
for i in range(10,20):
    print(f"i=={i}")

    if i==15:
        print(f"Value in i variable is {i}")
    print("*"*30)
```

```
i==10
******************************
i==11
******************************
i==12
******************************
i==13
******************************
i==14
******************************
i==15
Value in i variable is 15
******************************
i==16
******************************
i==17
******************************
i==18
******************************
i==19
******************************
```

In [20]:
```
for i in range(10,20):
    print(f"i=={i}")

    if i==15:
        print(f"Value in i variable is {i}")
        continue
    print("We are learning For loop")
    print("*"*30)
```

```
i==10
We are learning For loop
******************************
i==11
We are learning For loop
******************************
i==12
We are learning For loop
******************************
i==13
We are learning For loop
******************************
i==14
We are learning For loop
******************************
i==15
Value in i variable is 15
i==16
We are learning For loop
******************************
i==17
We are learning For loop
******************************
i==18
We are learning For loop
******************************
i==19
We are learning For loop
******************************
```

## 3. pass

In [21]:
```
for i in range(5):
    print(f"i=={i}")
```

```
i==0
i==1
i==2
i==3
i==4
```

In [23]:
```
for i in range(5):
```

```
  Cell In[23], line 1
    for i in range(5):
                      ^
SyntaxError: incomplete input
```

In [24]:
```
for i in range(5):
    pass
```

In [25]:
```
for i in range(10,20):
    print(f"i=={i}")

    if i==15:
        print(f"Value in variable is {i}")
        pass
    print("We are learning For loop")
    print("*"*30)
```

```
i==10
We are learning For loop
******************************
i==11
We are learning For loop
******************************
i==12
We are learning For loop
******************************
i==13
We are learning For loop
******************************
i==14
We are learning For loop
******************************
i==15
Value in variable is 15
We are learning For loop
******************************
i==16
We are learning For loop
******************************
i==17
We are learning For loop
******************************
i==18
We are learning For loop
******************************
i==19
We are learning For loop
******************************
```

### For -Else

In [26]:
```
for i in range(5,10):
    print(i)
```

```
5
6
7
8
9
```

In [27]:
```
for i in range(5,10):
    print(i)
else:
    print("All iterations are completed")
```

```
5
6
7
8
9
All iterations are completed
```

In [31]:
```
for i in range(5,10):
    print(f"i=={i}")
    if i==7:
        break
else:
    print("All iterations are completed")
```

```
i==5
i==6
i==7
```

In [32]:
```
for i in range(5,10):
    print(f"i=={i}")
    if i==7:
        continue
else:
    print("All iterations are completed")
```

```
i==5
i==6
i==7
i==8
i==9
All iterations are completed
```

In [ ]:
```

```